Title : ScholarTrack Pro Smart Attendance Companion

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Cluster : 1
Section : 4

Under the Guidance
of
Supervisor Name :

(DEEMED TO BE UNIVERSITY)

Abstract

•**Scholar Track** is a web-based attendance management system for students.

•Provides a **centralized dashboard** to monitor real-time attendance statistics.

•Allows **subject-wise attendance tracking and percentage calculation**.

•Includes an **attendance predictor** to estimate safe leave and required classes.

•Supports **custom attendance targets and personalized profiles**.

•Features a **responsive interface with dark/light mode and local data storage**.

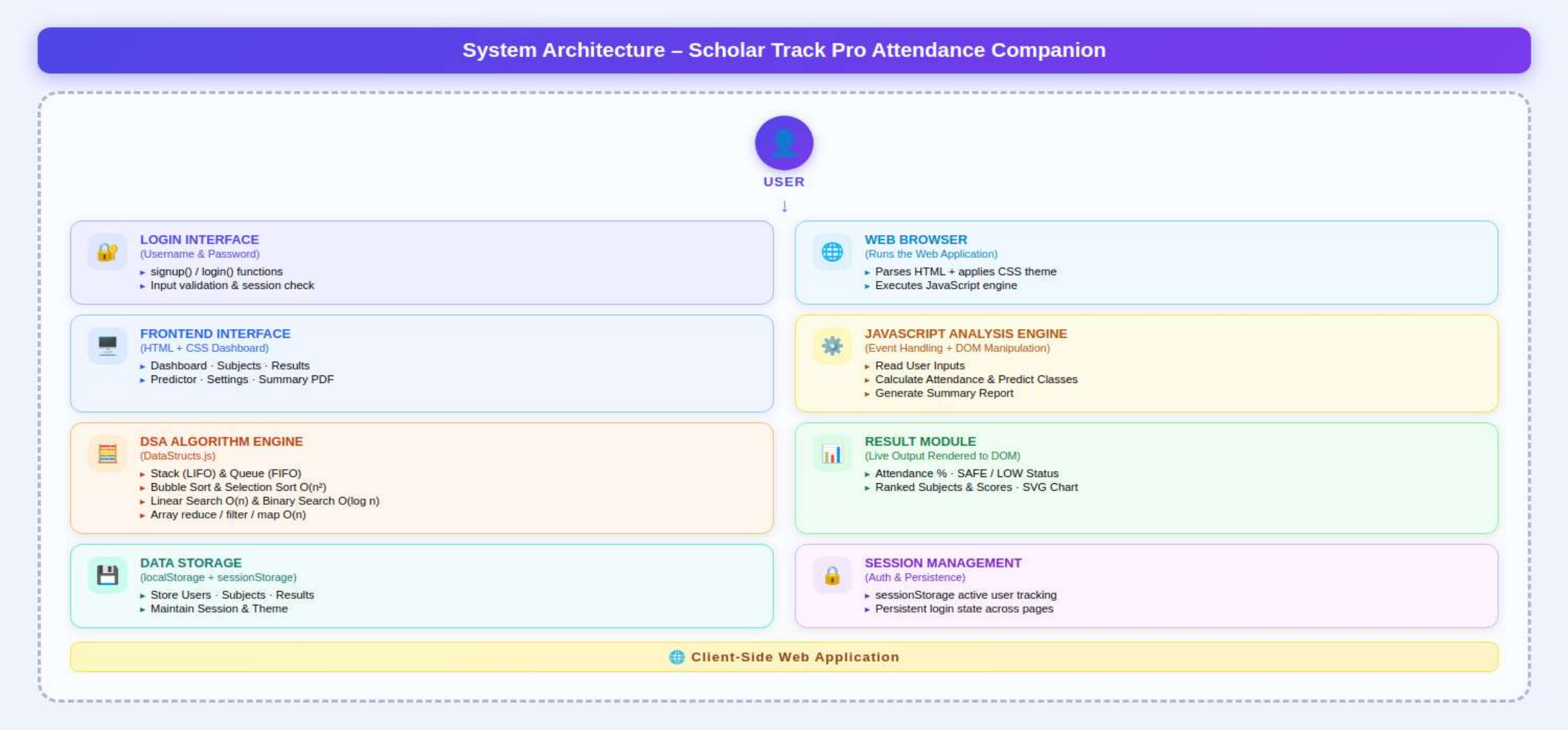
Existing System

- The traditional attendance system is mostly **manual or spreadsheet-based**, where records are maintained by teachers.
- Attendance is usually marked **by calling student names or using paper registers**, which takes more time.
- Calculating attendance percentages is often **done manually**, which may lead to errors.
- There is **no proper prediction system** to determine how many classes can be skipped safely.
- Attendance records may **be misplaced or difficult to access** when stored manually.
- Managing attendance for **multiple subjects becomes complex and time-consuming**.
- The existing system often **lacks a centralized digital platform** for students to monitor attendance.

Proposed System

- The proposed system is a **web-based Smart Attendance Management System** called **ScholarTrack Pro Smart Attendance Companion.**
- It provides a **centralized dashboard** to monitor overall attendance statistics.
- Students can **add and manage subjects** easily within the system.
- An **attendance predictor** helps students estimate how many classes they can skip or must attend.
- The system supports **custom attendance targets** based on institutional requirements.
- All attendance data is **stored digitally using browser local storage**, reducing paperwork.
- It offers a **responsive and user-friendly interface with dark/light theme support**.
- The system improves **accuracy, efficiency, and accessibility of attendance tracking**.

System Architecture



Project Modules

Login / User Profile Module

- Provides access for **students to log in and manage their profiles**.
- Allows users to **set personal attendance targets** and preferences.

Dashboard Module

- Displays **overall attendance statistics and summary**.
- Shows **subject-wise attendance percentage and performance**.

Subject Management Module

- Allows users to **add, edit, and delete subjects**.
- Stores subject details like **subject name and total classes conducted**.

Attendance Tracking Module

- Enables students to **mark present or absent for each class**.
- Automatically **updates attendance records and percentages**.

Attendance Prediction Module

- Predicts **how many classes can be skipped or must be attended**.
- Helps students **maintain required attendance percentage**.

Result Management Module

- Displays **semester results and subject-wise marks/grades**.
- Helps students **monitor academic performance along with attendance**.

Data Storage Module

- Stores attendance and result data using **browser local storage**.
- Ensures **quick and persistent access to records**.

Objectives

- To develop a **smart attendance tracking system** for students.
- To help students **monitor attendance percentage in real time**.
- To provide **subject-wise attendance analysis**.
- To assist students in **planning class attendance using prediction tools**.
- To reduce the risk of **attendance shortage before exams**.
- To create a **simple, user-friendly, and responsive web application**.

Co’s Attainments - FWD

- **CO1: Internet Fundamentals, HTML, Intro CSS**
- Basics of internet and web: HTTP/HTTPS, client-server model, browsers, hosting, domains.
- HTML structure and core content tags: document skeleton, headings, paragraphs, links.
- Intro CSS: syntax, selectors, typography, colors, and spacing fundamentals (margin, padding, border).
- **CO2: Forms, Semantic HTML, Advanced CSS Layout**
- HTML forms: form elements, input types/attributes, and validation basics.
- Structured content: semantic tags, tables, media elements, and accessibility attributes.
- CSS in depth: advanced selectors, Flexbox, Grid, block layout, media queries, transitions, animations, CSS functions, custom properties, Bootstrap utilities, SCSS basics, and responsive/themed UI design.
- **CO3: JavaScript Programming Essentials**
- Core JavaScript: operators, conditions, loops, functions, arrow functions, callbacks.
- Data structures and OOP basics: objects, inheritance concepts, arrays and array methods.
- Number formats: hexadecimal and scientific literals.

- **CO4: JavaScript Interactivity and DOM**
- DOM and event handling: selecting/updating elements and responding to user actions.
- Browser storage and async JS: local/session storage, asynchronous programming, and Promises.
- **CO5: Advanced Web Dev and Deployment**
- Error handling and ES6 modules.
- Dynamic forms and API integration with JavaScript.

Co’s Attainments - DSA

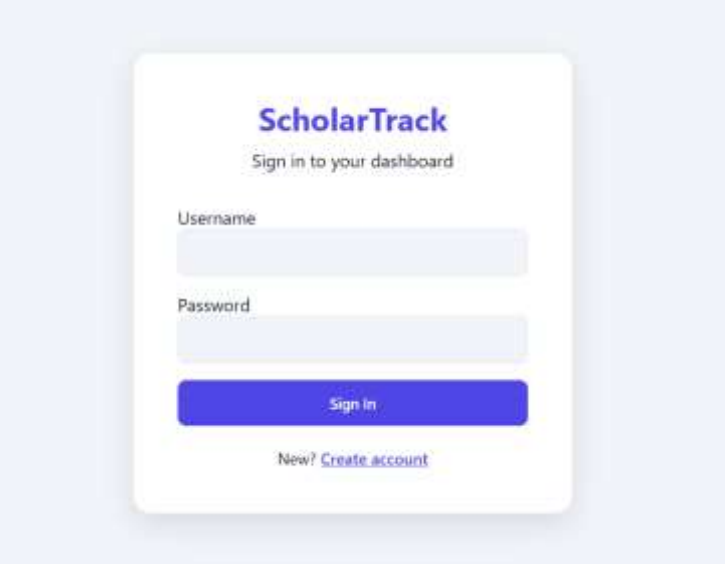
- **CO1: Algorithm Analysis, Searching, and Basic Sorting**
- Algorithm analysis foundations: mathematical background, computational model, and running-time calculations.
- Searching techniques: Linear Search and Binary Search with analysis.
- Sorting introduction with basic methods: Bubble Sort, Insertion Sort, Selection Sort.

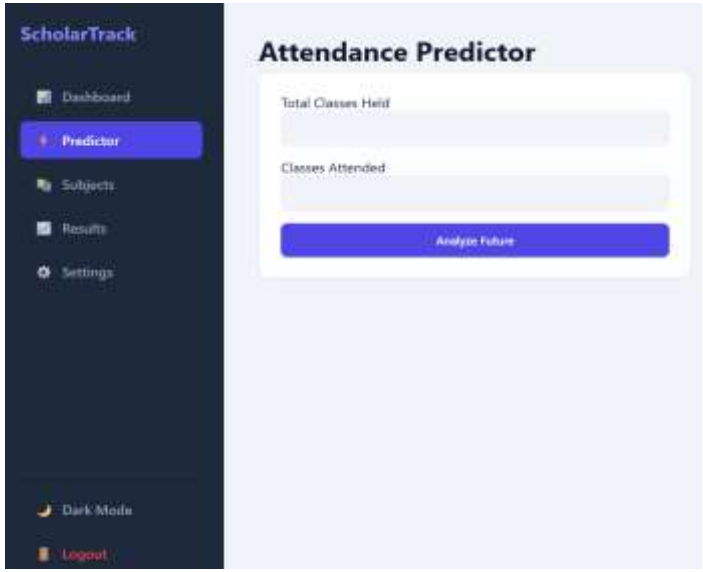
- **CO2: Advanced Sorting Techniques**
- Divide-and-conquer sorting methods: Merge Sort and Quick Sort.
- Comparative understanding of sorting algorithms based on time complexity and use cases.

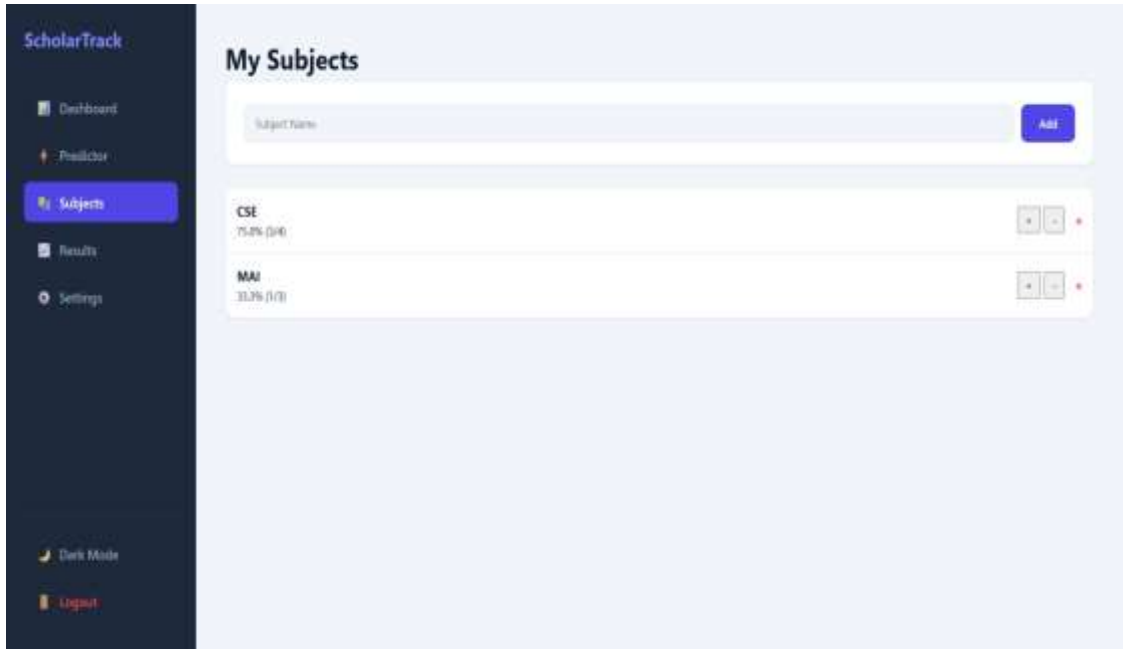
- **CO3: Lists and Linked List ADTs**
- Abstract Data Types (ADTs) and List ADT concepts.
- Linked list implementations: singly, doubly, and circular linked lists.
- Applications of lists and introduction to Polynomial ADT representation.

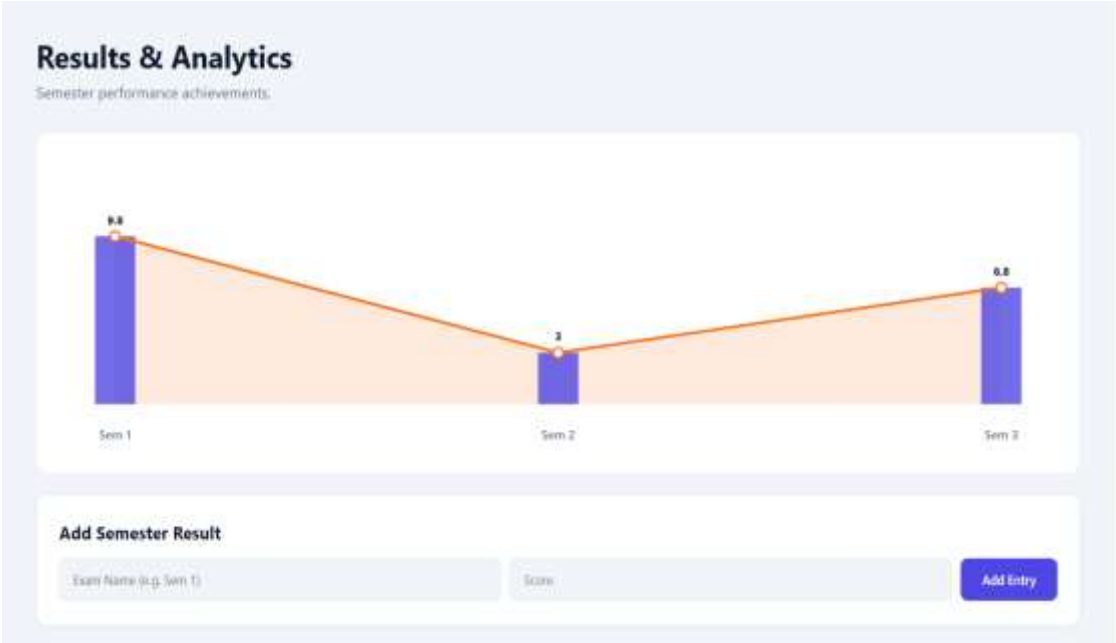
- **CO4: Stacks and Queues**
- Stack ADT: operations and applications.
- Expression-related applications: infix to postfix conversion, arithmetic expression evaluation, balancing symbols.
- Queue ADT: operations, Circular Queue, Dequeue (Deque).
- Implementing stacks and queues using linked lists.
- **CO5: Priority Queues, Hashing, and Trees-Related Structures**
- Priority Queues (Heaps): model, simple implementations, binary heap, and applications.
- Hashing: introduction, hash functions, collision handling through Separate Chaining and Open Addressing.
- Advanced hashing concepts: Rehashing and Extendible Hashing.

Screenshots for FWD & DSA









Conclusion

- The **ScholarTrack Pro Smart Attendance Companion** provides an efficient and user-friendly solution for managing academic attendance digitally.
- • The system allows students to **track subject-wise attendance, monitor performance, and maintain required attendance levels** through a centralized dashboard.

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- Overall, ScholarTrack simplifies attendance monitoring and supports students in **maintaining academic compliance effectively**.

Geo- Tag Photos

THANK YOU